

Internet of Things: Low Cost Monitoring BeeHive System using Wireless Sensor Network

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Abstract—Precision Beekeeping is an emerging research direction of Internet Things, based on a management strategy of individual monitoring of the colonies to minimize consumption resources and maximize the productivity of bee, is a challenging topic in the conditions of helping to add value to the apicultural chain, preserve the species, take care of the environment and contribute to the economic and social development of the territory. This document proposes an IoT system that can be applied into the hive to help beekeepers to get better control through precision measurements. The need and importance of how this system can collaborate to carry out more in-depth studies in the region through precision beekeeping are presented too. Therefore this document proposes a system architecture that can help to increase the productivity and accuracy using sensors to measure humidity, temperature, and weight remotely to finally send this data from the hive to the third-party online Ubidots platform services.

Keywords—*Internet of Things, Monitoring System, Precision Apiculture, Precision Beekeeping, Smart Apiary*

I. INTRODUCTION

Nowadays, the alarming situation that apiculture and bees suffer worldwide is mainly correlated to the fact that for extended periods, the evolution of plants and flowers was always conditioned by resistance and adaptation [1]. The human being in a few years has drastically modified the life cycles, as well as the environments on which they depend. In addition to this and other factors, bees can't adapt to this accelerated pace that has been imposed on them and consequently they are collapsing around the world [1]. Factors such as the transformation and radical alteration of forestry and agrarian landscapes, deforestation and intensive agriculture, the systematic multiplication of couriers, antennas and wireless networks present today in almost all natural environments, the strong tendency to monoculture and use of transgenic seeds in the fields around the world, the generalization and use of pesticides, insecticides and all kinds of agrochemicals to control pests or accelerating the growth of

crops have affected the sense of orientation, memory, and metabolism of bees and other insects [2].

Likewise, the increasing air pollution and gradual climate change is affecting and altering the flowering time of the plants and the rainy season, which harms the chemical messages that the flowers emit and as a result the bees lose orientation becoming more difficult to locate them, decreasing the quantity and quality of nectar, feeding poorly and with greater effort [2].

All these antecedents create a discouraging panorama leaving them defenseless with a weak immune system and a very sensitive metabolism which don't allow them to cope with pests or invaders that attack them and affect them directly and in a specific way. We are currently in a dangerous vicious circle, where they cannot find flowers and do not eat properly, in the meantime flowers don't reproduce when aren't pollinated because bees, flowers, fruits, and animals evolved at the same time tens of millions of years ago, the circle would only end with the collapse of that natural gear. Altering or modifying this equilibrium would destroy or condemn the existence of others. The bee family "Apis Mellifera" is one of the main species that collect nectar to turn it into honey and one of the most used by beekeepers around the world [3].

The research project presented in this document is focused on the collection, storage, and display of data in real-time of variables such as temperature, humidity, weight and noise, parameters that can be used for the study of behavior patterns or infer about states of the hive [4].

This document is organized as follow section two presents precision beekeeping term as well as the main factors involved in its proper performance, as well as describing the importance of the approach and the development of the system. Section 3 describes the study objects and opportunity areas that derive from the recording and analysis of the variables of temperature, humidity, weight, and sound. Section 4 presents a system developed hardware architecture and finally section 5 presents some results using the *ubidots* platform.

II. IOT IN PRECISION BEEKEEPING

A. *Precisión Beekeeping.*

Precision Beekeeping is an emerging research direction of the Internet of Things, based on a management strategy of individual monitoring of the colonies to minimize consumption resources and maximize the productivity of bees. Precision beekeeping is a challenging topic and in the conditions of helping to add value to the apicultural chain, preserves the species, take care of the environment and contribute to the economic and social development of the territory [4].

B. *Factors that intervene in the use of precision beekeeping*

The systems used in precision beekeeping currently present a combination of challenges that intervene in their use, acquisition and total integration in productive activities such as this [4][5]. Beekeeping should be developed collectively or in conjunction with other beekeepers to minimize expenses where additional value-added products (medicines, food products, cleaning products) should be generated to make the activity economically profitable [5].

C. *On the importance of the approach and development of the system.*

It's extremely important to distinguish the contributions that monitoring systems for precision beekeeping can give us. These systems should not be thought of in terms of automation or substitution of the labors of a beekeeper, activities such as checking the hive to determine the health state of the queen, the number of provisions, the health of the hive, the lack of space in the hive, breeding and raising camera among others are still activities that no beekeeper should stop doing since this would involve various associated risks [5].

Being that its true importance lies in terms of research, that is, a tool for collecting data on the different environmental variables of the hive during the entire growing season whose accuracy can contribute generate annual records. The precision data obtained should be compared to the empirical knowledge of years of experience of the beekeepers who later through observation and analysis get help to formulate, explain and check various phenomena and behaviors associated with bees. The availability of the information collected is of special interest to beekeepers, biologists, ethologists, and professionals who somehow carry out a deep research in beekeeping. That is why it is considered a critical factor to have this data current at any time at any place [5][6].

III. AREAS OF OPPORTUNITY AND OBJECTS OF STUDY

The monitoring methods currently focus on the measurement of variables such as weight, temperature, relative humidity in the air, concentration of gases in the hive, vibration and sound, parameters that can be constantly measured and shared in an accessible way with researchers, since the cost and size of electronic sensors decrease while their accuracy and capacity increase [7][8].

A. *Weight measurements on the hive*

Collecting continuous data on weight variations in the hive can help us determine or infer about:

- *The growth and consumption of food in the colony*
- *If there is nutritional stress due to a decrease in weight in the colony.*
- *The mass of nectar and pollen collected during the day and its relationship with the evaporation of water at night.*
- *The range of daily nectar collection once the nectar flow season begins.*
- *Swarming, that is, the natural way in which bees reproduce and form part of the colony.*
- *The start of the hibernation season.*
- *It is possible to estimate the number of foraging bees that monitor weight loss in the mornings when they go out to collect.*
- *Knowing the beginning and end of the data flow season can help explain how climate changes affect bees [7].*

There is research in the US by NASA's Goddard Space Center, a project that serves as a national observatory, where volunteer beekeepers from the Mid-Atlantic region weigh their colonies daily and manually feed the system with these data to understand the plant-pollinator relationship [8]. As a result, they obtained that the nectar flow season currently occurs 4 weeks earlier than in the year 1970 for the Maryland area [9][10].

B. *Temperature measurements in the colony*

Based on the temperature information, beekeepers and researchers have tried to detect conditions in the colony such as:

- *The increase in food consumption.*
- *The beginning of the breeding stage.*
- *The volume of breeding in the hive.*
- *Detection of pre-swarm states.*
- *Death in the bee colony.*
- *Behaviors associated during the pupal stage [10].*

Exposure to low temperatures during operculated breeding states induces high mortality and shortening in worker longevity. Exposure (20°C) during the operculated breeding states induces abnormal venation of the wings in the honeybee [6]. The worker bees keep the breeding nest of the colony within a narrow temperature range of 34 ± 1.5 °C; however, studies have shown that the worker bees that are raised during the state of pupa at 36 °C develop better capacities of learning and short-term memory [11]. Likewise, they develop better abilities to perform the communication dances, movements of the abdomen that are performed by the bees that inform the rest of the hive where the food source is located, indicating the direction and distance [11].

Why do the workers maintain the breeding temperature at $34 \pm 1.5^\circ\text{C}$ instead of 36°C if the temperatures slightly above normal improve the behavior associated with the collection of nectar and pollen in bees? One possibility is that incubation at a higher temperature than normal demands excessive energy from the colony, harming other physiological or neuronal factors that have not yet been proven, or that temperatures higher than $34 \pm 1.5^\circ\text{C}$ imply the generation of pathogens [12].

In the same way, temperature measurements on the upper part of the hive have been proposed for the recognition of the five periods of development of the annual cycle in the bee colony.

- *Beginning of winter breeding*
- *Beginning of spring breeding*
- *Beginning of summer breeding*
- *Beginning of autumn breeding*
- *Periods without breeding*

C. Moisture monitoring and atmospheric pressure in the colony

Based on the correlation of humidity, temperature, solar radiation, atmospheric pressure, rainfall, wind direction, and wind speed, models have been proposed to predict the foraging activity of honey bees using local weather conditions. As well as the correlation between temperature and humidity in the hive has been studied to investigate the relational metabolic and homeostasis activities [12].

IV. SYSTEM ARCHITECTURE

The temperature, humidity & weight will be read from the DHT11 and the load cells that are connected to the ESP8266. This transducer data can be accessed from anywhere connected to the internet, for means of configuration, programming or maintenance. Figure 1 shows the IoT architecture of the complete system, where the important function of the MQTT communication protocol can be observed, using this architecture arrangement it's possible to manage the incoming transducer data to display 24 hours of the day in a dashboard with the capability to visualize the information in any mobile device improving the ubiquitous service.

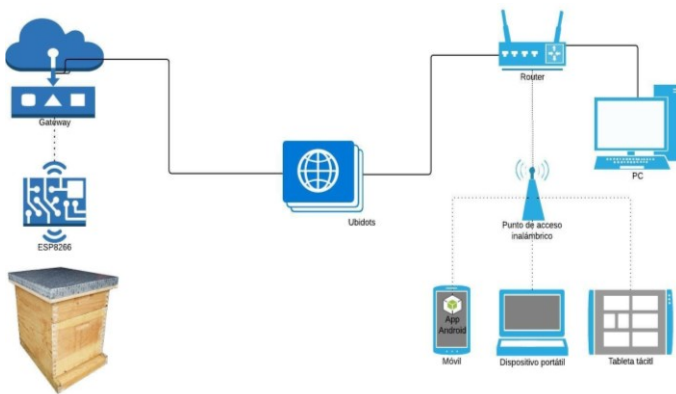


Fig. 1. Illustrative IoT architecture of the system.

A. System Architecture

The Figure 2 illustrates an ESP8266 which supports 802.11 b/g/n Wi-Fi protocol, integrated a low power 32 bit microcontroller a 10 bit analog to digital converter, with a 3.0 to 3.6v operating voltage and average operating current of 80mA, it includes a Tensilica Xtensa LX106 32 bit RISC CPU running at 80 MHz, with 16 GPIO Pins and SPI, I²C, I²S, UART – 2x TX and 1x RX interfaces, the operating temperature is from -40°C to 125°C and it has a frequency range 2400 to 2483.5MHz [13]. ESP8266 belongs to Espressif Systems' Smart Connectivity Platform, which is a set of high performance, high integration wireless SoC (System on Chip), intended for power and space-constrained mobile platform designers. It is a self-contained WIFI networking solution which means that we can store and run the application in it without the aid of any external processors, in the Figure 2 a pin-out diagram of ESP8266 is presented [14][15].

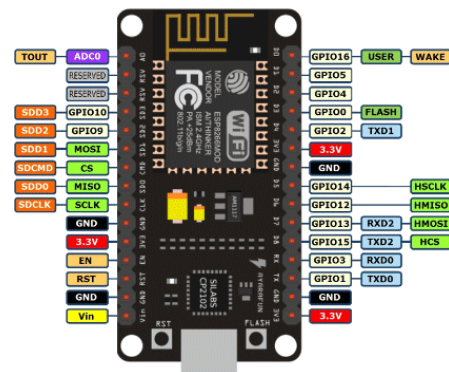


Fig. 2. Node ESP8266.

To measure temperature and humidity there are a DHT11 and DS18B20. The main reason behind to use these two options is that an arrangement of four DS18B20 can be mounted on the wall of each honey supers when the bee population increments and demands grow. Figure 3 shows the structure of the *langstroth* hive.

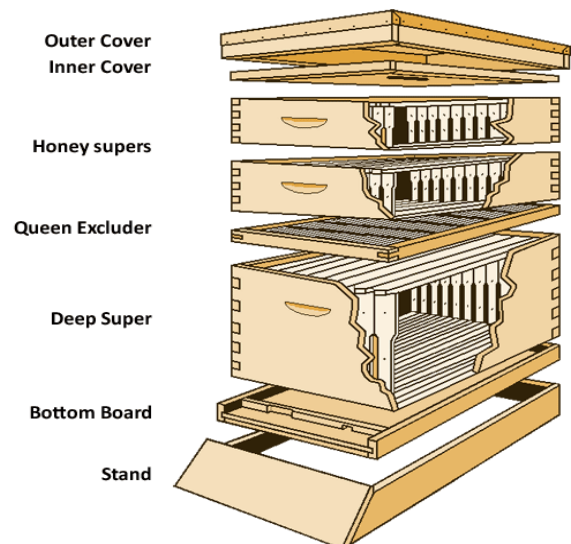


Fig. 3. Langstroth hive.

An arrangement of four 50 kg load cells are used in conjunction with a Load Cell Transmitter Hx711, it's well known by the experience of beekeepers that the weight of hive oscillates from 50kg to 200 kg that's the main reason for use 4 load cells. Figure 4 shows an item that helps to the integration of the load cells into the hive it's a 3D printed model with screws support.

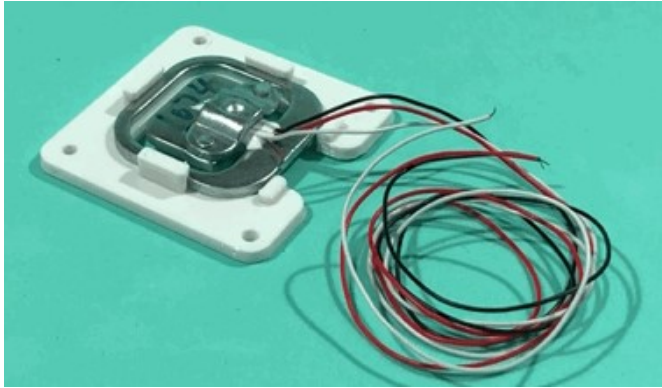


Fig. 4. Load Cell and 3D model support.

The frequency of data transmission uses the time interval approaches since the application for this system requires the acquisition of data every 5 seconds from the initialization of the system. The communication standard used for communication between the gateway and *Ubidots* is JSON (JavaScript Object Notation) which was defined specifically for sensor networks and is a message protocol that aims to connect embedded devices and networks with applications and middleware, which uses the publish/subscribe communication model to provide transitional flexibility and simplicity of implementation and is suitable for devices with limited resources that use unreliable or low bandwidth links[10]. A block diagram from the embedded hardware architecture is shown in the Figure 5.

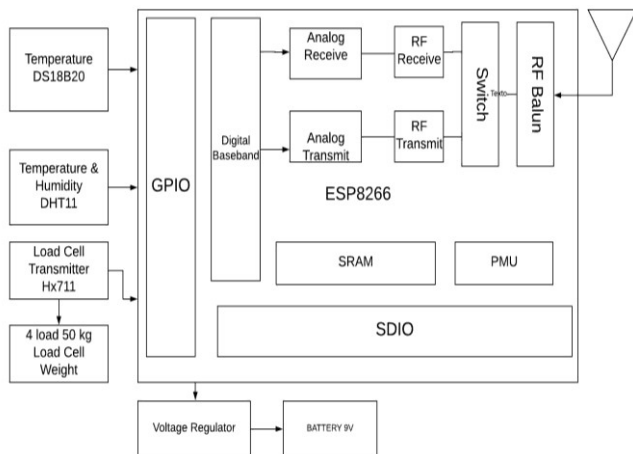


Fig. 5. Block diagram

Finally *Ubidots* is a cloud service that allows us to store and interpret information from sensors in real-time, making it possible to create applications for the Internet of Things in an easy and fast way, saving time and money when developing

applications such as GPS telemetry systems, systems for temperature monitoring, applications for counting vehicles in a street, etc. A great advantage of *Ubidots* is that it offers a free plan, with which prototypes and 100% functional applications are carried out [11]. The wiring diagram to the embedded system is shown in Figure 6.

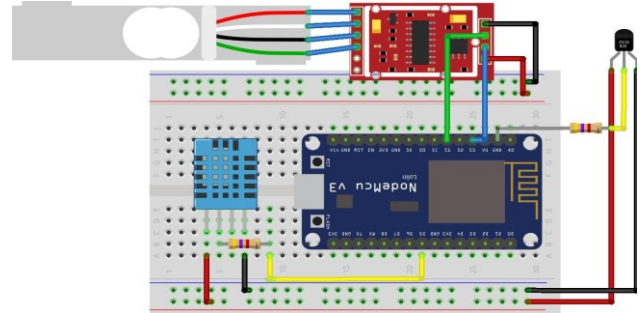


Fig. 6. Wiring diagram.

Figure 7 shows a block diagram about the implemented algorithm, first all de I/O pins are set up like a inputs where the transducers are connected, the data incoming by the DHT11 and the load cells are processed by the ESP8666 and then are sent via the MQTT protocol to the *ubidots* web server where a dashboard displays the variables about de data transducers collected, at the same way some alerts are setup trough *ubidots* IoT Platform.

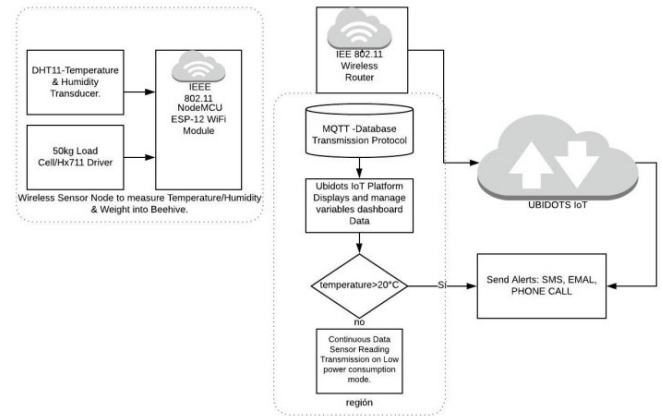


Fig. 7. Block diagram of implement algorithm.

B. About IoT Architecture.

Every time a device updates a sensor value in a variable, a data-point or "dot" is created. *ubidots* stores dots that come from devices inside variables and these stored dots have corresponding timestamps.

This layer digitizes and transfers the data to the object abstraction layer through secure channels, the data here is transferred through the Wi-Fi protocol where the data management process is carried out [16][17].

The object abstraction layer transfers the data produced by the objects to the service management layer where decisions are made and deliver services to those who request it through the network protocols. The application layer is the interface

through which the end-users can interact with the device; this layer provides the services, that is, the measurements to the client that requests the data [18]. Finally, the business layer manages all the activities and services of the IoT systems.

It is responsibilities to build a business model, graphs, flow diagrams based on the data received from the application layer, making it possible to support decision-making processes based on Big Data analysis, as well as monitoring the 4 layers underlying, compares the output of each layer with the expected output to improve services and maintain user privacy [19][20].

V. COST OF THE PROTOTYPE

An important aspect of this prototype is that it is considered low cost. The estimated cost of this project is shown below in Table 1 where every single part and its cost are listed. Therefore, the total estimated cost would be about \$17.49 USD. Compared to other monitoring systems in the market it is cheaper, a market comparison is shown in Table 2.

TABLE I. ESTIMATED COST OF THE PROTOTYPE

MATERIAL	QUANTITY	COST (USD)
ESP8266	1	\$4.00
DHT11 (temperature and humidity)	1	\$1.30
DS18B20 (temperature)	1	\$1.20
DC power source	1	\$2.00
50kg Load Cell & Half Bridge/Amplifier HX711 (weight)	4	\$8.99
Total		\$17.49

TABLE II. COMPARISON OF PRODUCTS

PRODUCT	COST (USD)
Hive heart, without battery (temperature, and humidity)	\$55.00 USD
Hive Scale, without battery (temperature, and humidity)	\$100.46 USD
BuzzBox Mini (temperature, and humidity)	\$199.00 USD
HM-5 Hive Monitor (temperature and weight)	\$290.00 USD
Prototype (temperature, humidity and weight)	\$17.49 USD

VI. RESULTS

The following pictures show the results of the instrumented beehive. First, the load cells were adjusted to the Bottom board this part of the hive could be seen as the floor of the beehive it consists of several rails that serve as a frame around a solid piece of wood and it protects the colony from the damp ground. These days, more and more beekeepers are using what's called a *screened* bottom board in place of the standard bottom board. This improves ventilation and is helpful when controlling and monitoring the colony's population of *varroa mites*, the complete instrumented bottom board is shown in Figure 9.

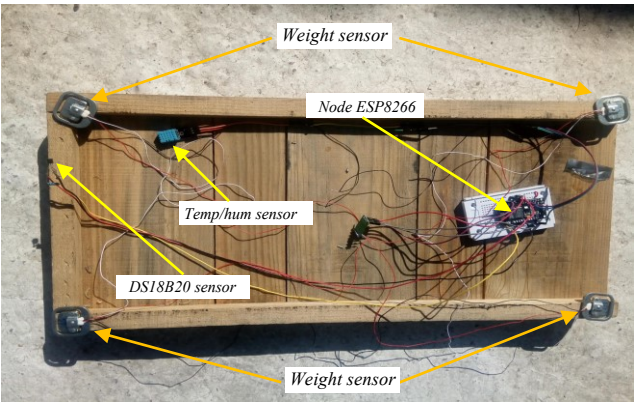


Fig. 9. Instrumented bottom board.

Since the deep super is mounted on the top of the bottom board the first idea was put the four load cells in each corner, take the weight from the deep super helped us to run some calibrating test. Figure 10 a) shows the bottom and deep super assembly and Figure 10 b) shows the final mounting system.

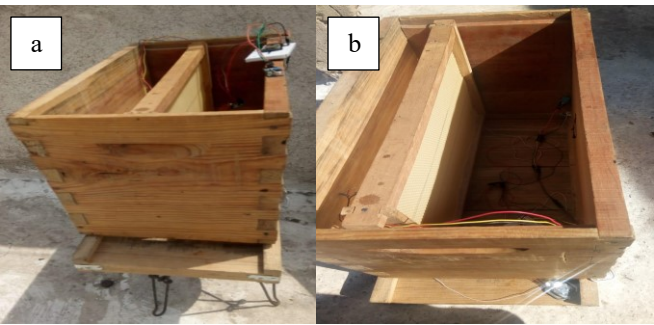


Fig. 10. a) Bottom and deep super assembly and b) final mounting system.

Figure 11 shows a real-time graphics corresponding to the temperature variable, showing the temperature value, the date and the time of recording.



Fig. 11. Data logger.

Ubidots let us create static and dynamic dashboards, user interfaces to organize and present a device's data and the insights derived from the data. The dashboards use widgets that display the data as charts, indicators, controls, tables, and graphs. The dashboard generated in the platform uses thermometers widgets to visually show their records in Celsius

and Fahrenheit degrees incoming data from DHT11 and DS18B20 transducers. Also, a histogram and a database that can be useful for various research analysis purposes were generated too. Figure 12 shows the dashboard arrangement to the real-time temperature and humidity readings. Also, a real-time database and histogram widgets were generated too and are shown in Figure 12.

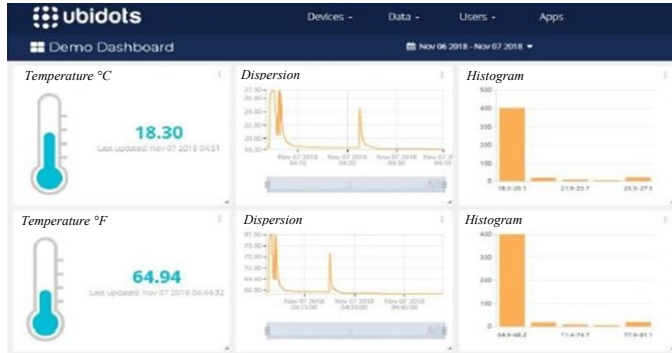


Fig. 12. DHT11 temperature and humidity dashboard.

The temperature was incoming from the DS18B20 transducer and the weight measure through the 4 load cells are displayed too, presented in an easy visualization way to facilitate the work of the beekeeper, a numeric representation widget is used to present the weight readings. Figure 13 shows the dashboard arrangement to the weight and temperature of real-time readings.

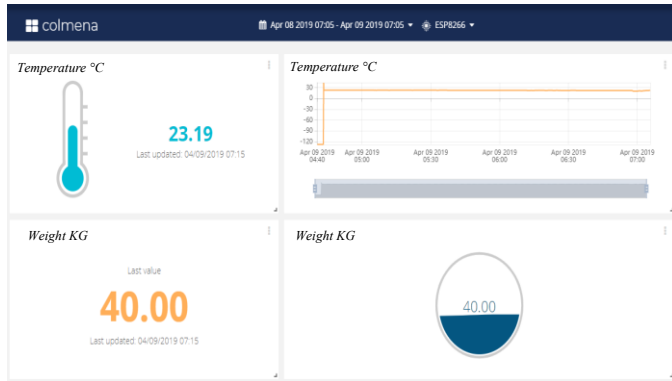


Fig. 13. Temperature and weight dashboard.

An alert was developed based on the temperature variability which has the objective in the first instance of measuring the temperature in the *operculated breeding* states.

Since it is known that the exposure to low temperatures during the *operculated breeding* states induces high mortality and shortening in the longevity of the workers, the exposure (20°C) during the *operculated breeding* states induce the abnormal venation of the wings in the honey bee.

To do this, the first thing was to set the alarm so that when the threshold of 20°C was exceeded, a voice call, an email and an SMS were sent to the beekeeper, who could take the appropriate precautionary actions.

The Figure 14 a) shows the email that was sent once the temperature passed the threshold of 20°C and the alert was triggered. Figure 14 b) shows the SMS text message that was sent once the temperature above passed the threshold of 20°C and the alert was triggered.

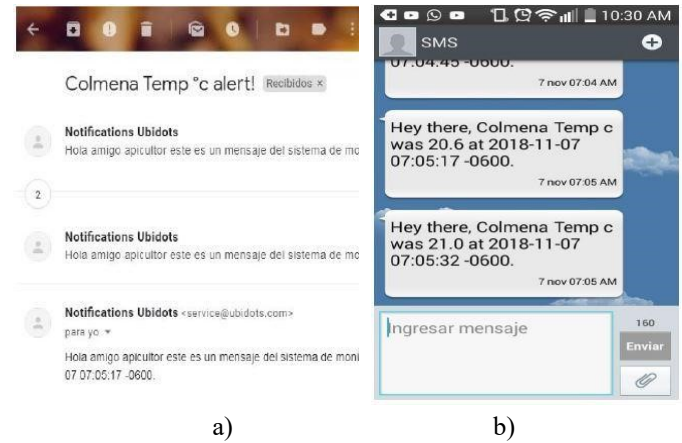


Fig. 14. a) Email notification, and b) SMS notification.

VII. CONCLUSION

Precision beekeeping is a research approach or direction which may be considered vital to carry out environmental records with greater accuracy, hard data obtained through electronic transducers which can be inserted in the hive in order to generate bases of data that once reviewed by specialists in the area and compared with the empirical experience of beekeepers may result in a greater understanding of the relationship between the territory's flora and fauna and establishing patterns of behavior in the family of the "Apis mellifera", on this research we explore some possibilities that an IoT hive can carry out to develop more in-depth studies, the prototype developed has some opportunity areas that will developed in the future like the hardware full integration to the hive structure, its well-known that an a beekeeper requires frequently to disassemble the hive especially when a bee population increment exist so one problem is that incrust all the transducers on the same place does not originate precise measurements due to the limited sensing area especially when it's necessary to add honey supers, wiring all the hive results in an obstacle to the driving beekeepers technique.

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